

Answer Key & Marking Scheme

Subject: Science | **Test:** Test 1 (Day 7, 16 May 2026) — Weekly
Maximum Marks: 80

Syllabus: Chemical Reactions and Equations • Acids, Bases and Salts • Metals and Non-metals

SECTION A — Answer Key (1 mark each)

- Q1.** (c) Displacement — Fe is more reactive than Cu and displaces it from its salt.
- Q2.** (b) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$.
- Q3.** (a) adding antioxidants.
- Q4.** (c) grey.
- Q5.** (b) exothermic reaction — releases energy in the form of heat.
- Q6.** (c) 7.
- Q7.** (b) citric acid.
- Q8.** (b) dry slaked lime — $\text{Ca}(\text{OH})_2 + \text{Cl}_2 \rightarrow \text{CaOCl}_2 + \text{H}_2\text{O}$.
- Q9.** (b) Baking soda — NaHCO_3 has no water of crystallisation.
- Q10.** (d) 10 — pH > 7 is basic.
- Q11.** (c) Sodium.
- Q12.** (a) Mercury.
- Q13.** (d) $\text{AgNO}_3 + \text{Cu}$ — Cu is more reactive than Ag.
- Q14.** (b) Hydrogen.
- Q15.** (c) graphite.
- Q16.** (a) $\text{HCl} + \text{HNO}_3$ in the ratio 3 : 1.
- Q17.** Corrosion is the slow chemical attack of metals by air, moisture, or chemicals in the environment, leading to deterioration (e.g. rusting of iron).
- Q18.** Sodium reacts vigorously with the oxygen and moisture of air; kerosene prevents contact with both, so sodium is stored under it.
- Q19.** Hydrogen gas (H_2).
- Q20.** $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ (calcium sulphate hemihydrate).

SECTION B — Answer Key (2 marks each)

- Q21.** $2\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$. Type: double-displacement / neutralisation reaction. (2)
- Q22.** Respiration releases energy when food (glucose) reacts with oxygen, hence it is exothermic. $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Energy}$. (2)
- Q23.** pH is the negative logarithm (to base 10) of hydrogen-ion concentration; it measures acidity (pH < 7) or basicity (pH > 7). When pH in the mouth falls below 5.5, tooth enamel (calcium phosphate) starts dissolving, causing decay. (2)
- Q24.** Dry HCl gas does not produce H^+ ions. Only when HCl dissolves in water does it dissociate to give H^+ ions, which are responsible for acidic behaviour. (2)
- Q25.** An amphoteric oxide reacts with both acids and bases to produce salt and water. Example: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$; $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$. (2)
- Q26.** Ionic compounds have strong electrostatic forces between oppositely-charged ions in a closely-packed lattice. Breaking this lattice requires a large amount of energy, hence they are solids with high melting/boiling points. (2)

SECTION C — Answer Key (3 marks each)

Q27. Combination: two or more substances combine to form a single product. Example: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$. **Decomposition:** a single substance breaks into two or more simpler substances. Example: $2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$ (on heating). (3)

Q28. A redox reaction involves simultaneous oxidation and reduction. Example: $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$. Here CuO loses oxygen (reduced); H_2 gains oxygen (oxidised). H_2 is the reducing agent and CuO the oxidising agent. (3)

Q29. (i) **Baking soda** — Sodium hydrogen carbonate (NaHCO_3); used as an antacid. (ii) **Washing soda** — Sodium carbonate decahydrate ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$); used in cleaning and as a water softener. (iii) **Bleaching powder** — Calcium oxychloride (CaOCl_2); used for bleaching cotton/paper and disinfecting drinking water. (3)

Q30. (i) $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ (neutralisation). (ii) $\text{ZnO} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{O}$ (acid–base; ZnO is amphoteric). (iii) $\text{CO}_2 + \text{Ca(OH)}_2 \rightarrow \text{CaCO}_3 \downarrow + \text{H}_2\text{O}$ (lime water turns milky). (3)

Q31. Any three: (i) Metals are lustrous; non-metals are dull (except iodine, graphite). (ii) Metals are good conductors of heat and electricity; non-metals generally are not (graphite excepted). (iii) Metals are malleable and ductile; non-metals are brittle (in the solid state). (3)

Q32. Metals high in the activity series (K, Na, Ca, Mg, Al) are obtained by electrolytic reduction of their molten salts. Metals in the middle (Zn, Fe, Pb, Cu) are obtained by reducing their oxides with a more reactive substance (typically carbon). Metals low in the series (Hg, Ag, Au) are often found native or are obtained by simple thermal decomposition of their oxides/sulphides. (3)

Q33. Copper lies below hydrogen in the activity series, so it cannot displace H from acids. Iron lies above H, so it does react: $\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2 \uparrow$. (3)

SECTION D — Answer Key (5 marks each)

Q34. (a) (i) **Oxidation** — addition of oxygen / loss of hydrogen / loss of electrons. Example: $2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$. (ii) **Reduction** — loss of oxygen / gain of hydrogen / gain of electrons. Example: $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$. (iii) **Corrosion** — gradual deterioration of a metal due to attack by air/moisture/acids. Example: rusting of iron, $4\text{Fe} + 3\text{O}_2 + x\text{H}_2\text{O} \rightarrow 2\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$. (3)

(b) Painting forms a barrier between the iron surface and oxygen/moisture, preventing oxidation, hence rust does not form. (1)

(c) **Advantage:** the zinc coating sacrifices itself and protects iron even if scratched. **Disadvantage:** zinc layer wears off with time and may be expensive for large-scale items. (1)

Q35. (a) **Neutralisation:** reaction between acid and base producing salt and water. Example: $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$. (1)

(b) HCl in the stomach helps digestion (activates pepsin) and kills bacteria. Excess HCl causes acidity / heartburn. **Antacid:** milk of magnesia (Mg(OH)_2) or sodium hydrogen carbonate. (2)

(c) Common salt (NaCl) is the raw material for: (i) sodium hydroxide (chlor-alkali process) — $2\text{NaCl(aq)} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH(aq)} + \text{Cl}_2 \uparrow + \text{H}_2 \uparrow$; (ii) baking soda; (iii) washing soda; (iv) bleaching powder. (2)

Q36. (a) Activity series ranks metals by decreasing reactivity. Order: **Mg > Zn > Fe > Cu > Au**. (2)

(b) Tin is less reactive than zinc and does not react with food acids; if zinc were used, it would react with foodstuffs. (1)

(c) (i) $\text{Mg} + \text{H}_2\text{O(steam)} \rightarrow \text{MgO} + \text{H}_2 \uparrow$; (ii) $2\text{Fe} + 3\text{Cl}_2 \rightarrow 2\text{FeCl}_3$. (2)

SECTION E — Answer Key (4 marks each)

Q37. (i) $\text{Pb(NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 \downarrow + 2\text{KNO}_3$. (1) (ii) Double displacement / precipitation reaction. (1) (iii) Lead(II) iodide (PbI_2). (1) (iv) The ions exchange partners and an insoluble lead iodide precipitates out, removing them from solution and driving the reaction forward. (1)

Q38. (i) Chilli juice contains methanoic acid; baking soda is mildly basic and neutralises it, relieving the burn. (1) (ii) Tea contains tannins; the citric acid in lemon juice reacts with these and dissolves the stain. (1) (iii) Sodium hydrogen carbonate, NaHCO_3 . (1) (iv) Used as an antacid / in baking powder / in fire extinguishers. (1)

Q39. (i) Pure gold is very soft and gets deformed easily; alloying makes it hard. (1) (ii) Silver and copper. (1) (iii) 22-carat means 22 parts of gold by weight in 24 parts of the alloy (~91.7% pure gold). (1) (iv) An alloy is a homogeneous mixture of two or more metals (or a metal and a non-metal) with metallic properties. Example: brass ($\text{Cu} + \text{Zn}$) or steel ($\text{Fe} + \text{C}$). (1)

— End of Answer Key —